

CS-GY 9223 I: Lecture 10

Krylov methods, spectral clustering, spectral graph theory.

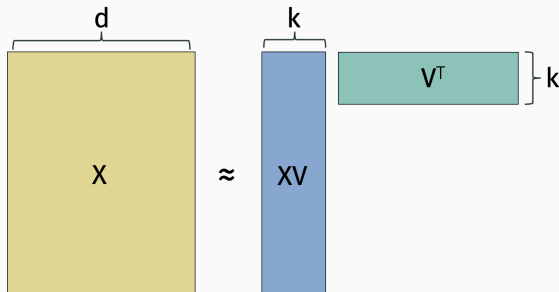
NYU Tandon School of Engineering, Prof. Christopher Musco

Three classes of methods.

- Direct Methods:
- Iterative Methods:
- Randomized Methods:

LOW-RANK APPROXIMATION

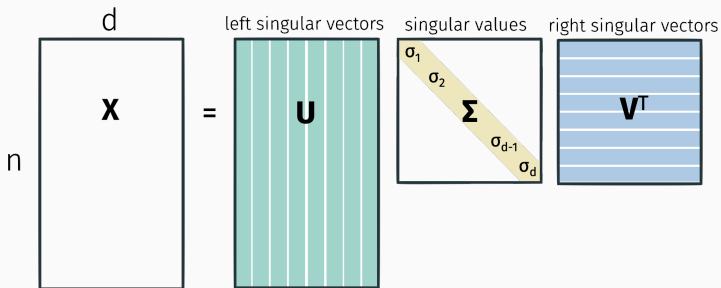
Write $\mathbf{X}$ as a rank k factorization by projecting onto the subspace spanned by an orthonormal matrix $\mathbf{V} \in \mathbb{R}^{d \times k}$



SINGULAR VALUE DECOMPOSITION

One-stop shop for computing optimal low-rank approximations.

Any matrix X can be written:



Where $U^T U = I$, $V^T V = I$, and $\sigma_1 \geq \sigma_2 \geq \dots \sigma_d \geq 0$.

COMPUTATIONAL QUESTION

Given a subspace $\mathcal{V}$ spanned by the k columns in $\mathbf{V}$,

$$\|\mathbf{X} - \mathbf{X}\mathbf{V}\mathbf{V}^T\|_F^2 = \min_{\mathbf{C}} \|\mathbf{X} - \mathbf{C}\mathbf{V}^T\|_F^2$$

We want to find the best $\mathbf{V} \in \mathbb{R}^{d \times k}$:

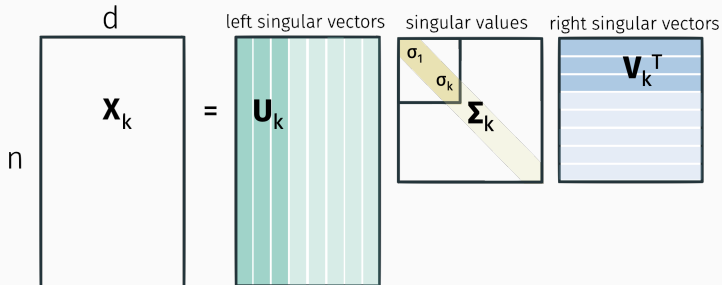
$$\min_{\text{orthonormal } \mathbf{V} \in \mathbb{R}^{d \times k}} \|\mathbf{X} - \mathbf{X}\mathbf{V}\mathbf{V}^T\|_F^2 \quad (1)$$

Note that $\|\mathbf{X} - \mathbf{X}\mathbf{V}\mathbf{V}^T\|_F^2 = \|\mathbf{X}\|_F^2 - \|\mathbf{X}\mathbf{V}\mathbf{V}^T\|_F^2$ for all orthonormal $\mathbf{V}$ (since $\mathbf{V}\mathbf{V}^T$ is a projection). Equivalent form:

$$\max_{\text{orthonormal } \mathbf{V} \in \mathbb{R}^{d \times k}} \|\mathbf{X}\mathbf{V}\mathbf{V}^T\|_F^2 = \|\mathbf{X}\mathbf{V}\|_F^2 \quad (2)$$

SINGULAR VALUE DECOMPOSITION

Can read off optimal low-rank approximations from the SVD:



$$\mathbf{X}_k = \mathbf{U}_k \mathbf{U}_k^T \mathbf{X} = \mathbf{X} \mathbf{V}_k \mathbf{V}_k^T.$$

$$\mathbf{V}_k = \arg \min_{\text{orthonormal } \mathbf{V} \in \mathbb{R}^{d \times k}} \|\mathbf{X} - \mathbf{X} \mathbf{V} \mathbf{V}^T\|_F^2 = \arg \max_{\text{orthonormal } \mathbf{V} \in \mathbb{R}^{d \times k}} \|\mathbf{X} \mathbf{V} \mathbf{V}^T\|_F^2$$

Today: What about when $k = 1$?

Goal: Find some $\mathbf{z} \approx \mathbf{v}_1$.

Input: $\mathbf{X} \in \mathbb{R}^{n \times d}$ with SVD $\mathbf{U}\mathbf{\Sigma}\mathbf{V}$.

Power method:

- Choose $\mathbf{z}^{(0)}$ randomly. E.g. $\mathbf{z}_0 \sim \mathcal{N}(0, 1)$.
- For $i = 1, \dots, T$
 - $\mathbf{z}^{(i)} = \mathbf{X}^T \cdot (\mathbf{X}\mathbf{z}^{(i-1)})$
 - $n_i = \|\mathbf{z}^{(i)}\|_2$
 - $\mathbf{z}^{(i)} = \mathbf{z}^{(i)} / n_i$

Return $\mathbf{z}_T$

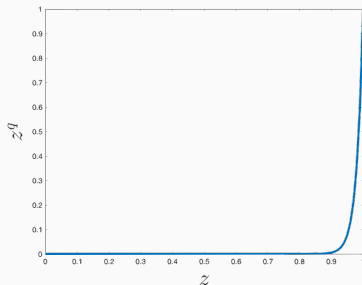
Theorem (Power Method Convergence)

Let $\gamma = \frac{\sigma_1 - \sigma_2}{\sigma_1}$ be parameter capturing the “gap” between the first and second largest singular values. If Power Method is initialized with a random Gaussian vector then, with high probability, after $T = O\left(\frac{\log d/\epsilon}{\gamma}\right)$ steps, we have:

$$\|\mathbf{v}_1 - \mathbf{z}^{(T)}\|_2 \leq \epsilon.$$

Total runtime: $O(T \cdot \text{nnz}(\mathbf{X})) \leq O(T \cdot nd)$

$$\mathbf{z}^{(q)} = \mathbf{c} \cdot (\mathbf{X}^T \mathbf{X})^q \cdot \mathbf{g}$$



$$\mathbf{z}^{(q)} = \mathbf{c} \cdot \left[c_1 \cdot \sigma_1^{2q} \mathbf{v}_1 + c_2 \cdot \sigma_2^{2q} \mathbf{v}_2 + \dots + c_n \cdot \sigma_n^{2q} \mathbf{v}_n \right]$$

$$\mathbf{z}^{(q)} = c \cdot (\mathbf{X}^T \mathbf{X})^q \cdot \mathbf{g}$$

Along the way we computed:

$$\mathcal{K}_q = \left[\mathbf{g}, (\mathbf{X}^T \mathbf{X}) \cdot \mathbf{g}, (\mathbf{X}^T \mathbf{X})^2 \cdot \mathbf{g}, \dots, (\mathbf{X}^T \mathbf{X})^q \cdot \mathbf{g} \right]$$

$\mathcal{K}$ is called the Krylov subspace of degree q .

Idea behind Krylov methods: Don't throw away everything before $(\mathbf{X}^T \mathbf{X})^q \cdot \mathbf{g}$. What you're using when you run `svds` or `eigs` in MATLAB or Python.

Want to find $\mathbf{v}$, which minimizes $\|\mathbf{X} - \mathbf{X}\mathbf{v}\mathbf{v}^T\|_F^2$.

Lanczos method:

- Let $\mathbf{Q} \in \mathbb{R}^{d \times k}$ be an orthonormal span for the vectors in $\mathcal{K}$.
- Solve $\min_{\mathbf{v}=\mathbf{Q}\mathbf{w}} \|\mathbf{X} - \mathbf{X}\mathbf{v}\mathbf{v}^T\|_F^2$.
 - Find best vector in the Krylov subspace, instead of just using last vector.
 - Can be done in $O(\text{nnz}(\mathbf{X}) \cdot k + dk^2)$ time.

Claim 1: For any degree q polynomial p , we can write $p(X^T X) \cdot \mathbf{g}$ as $\mathbf{Q}\mathbf{w}$ for some $\mathbf{w}$.

Claim 2:

$$\min_{\mathbf{v}=\mathbf{Q}\mathbf{w}} \|\mathbf{X} - \mathbf{X}\mathbf{v}\mathbf{v}^T\|_F^2 = \min_{\text{degree } q \text{ polynomial } p} \|\mathbf{X} - \mathbf{X}\mathbf{v}_p\mathbf{v}_p^T\|_F^2$$

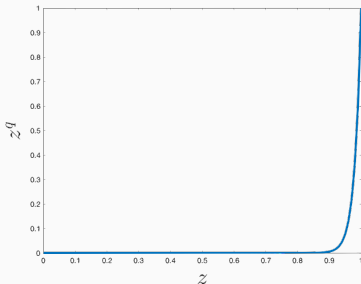
where $\mathbf{v}_p = p(X^T X) \cdot \mathbf{g}$.

Claim 3:

$$\mathbf{z}^{(q)} = c \cdot [c_1 \cdot p(\sigma_1^2)\mathbf{v}_1 + c_2 \cdot p(\sigma_2^2)\mathbf{v}_2 + \dots + c_n \cdot p(\sigma_n^2)\mathbf{v}_n]$$

LANCZOS METHOD ANALYSIS

Claim: There is an $O\left(\sqrt{q \log \frac{1}{\epsilon}}\right)$ degree polynomial $\hat{p}$ approximating $\mathbf{x}^q$ up to error ϵ on $[0, \sigma_1^2]$.



$$\|\mathbf{X} - \mathbf{X}\mathbf{v}_{p^*}\mathbf{v}_{p^*}^T\|_F^2 \leq \|\mathbf{X} - \mathbf{X}\mathbf{v}_{\hat{p}}\mathbf{v}_{\hat{p}}^T\|_F^2 \approx \|\mathbf{X} - \mathbf{X}\mathbf{v}_{x^q}\mathbf{v}_{x^q}^T\|_F^2 \approx \|\mathbf{X} - \mathbf{X}\mathbf{v}_1\mathbf{v}_1^T\|_F^2$$

Runtime: $O\left(\frac{\log(d/\epsilon)}{\sqrt{\gamma}} \cdot \text{nnz}(\mathbf{X})\right)$ vs. $O\left(\frac{\log(d/\epsilon)}{\gamma} \cdot \text{nnz}(\mathbf{X})\right)$

POWER METHOD – NO GAP DEPENDENCE

Convergence is slow when $\gamma = \frac{\sigma_1 - \sigma_2}{\sigma_1}$ is small. $\mathbf{z}^{(q)}$ has large components of both $\mathbf{v}_1$ and $\mathbf{v}_2$. But in this case:

$$\|\mathbf{X} - \mathbf{X}\mathbf{v}_1\mathbf{v}_1^T\|_F^2 = \sum_{i \neq 1} \sigma_i^2 \approx \sum_{i \neq 2} \sigma_i^2 = \sigma_i^2 \|\mathbf{X} - \mathbf{X}\mathbf{v}_2\mathbf{v}_2^T\|_F^2.$$

So we don't care! Either $\mathbf{v}_1$ or $\mathbf{v}_2$ give good rank-1 approximations.

Claim: To achieve

$$\|\mathbf{X} - \mathbf{X}\mathbf{z}\mathbf{z}^T\|_F^2 \leq (1 + \epsilon) \|\mathbf{X} - \mathbf{X}\mathbf{v}_1\mathbf{v}_1^T\|_F^2$$

we need $O\left(\frac{\log(d/\epsilon)}{\epsilon}\right)$ power method iterations or $O\left(\frac{\log(d/\epsilon)}{\sqrt{\epsilon}}\right)$ Lanczos iterations.

GENERALIZATIONS TO LARGER k

- Block Power Method aka Simultaneous Iteration aka Subspace Iteration aka Orthogonal Iteration
- Block Krylov methods
- Let $\mathbf{G} \in \mathbb{R}^{d \times k}$ be a random Gaussian matrix.
- $\mathcal{K}_q = \left[\mathbf{G}, (\mathbf{X}^T \mathbf{X}) \cdot \mathbf{G}, (\mathbf{X}^T \mathbf{X})^2 \cdot \mathbf{G}, \dots, (\mathbf{X}^T \mathbf{X})^q \cdot \mathbf{G} \right]$

Runtime: $O\left(\text{nnz}(\mathbf{X}) \cdot k \cdot \frac{\log d / \epsilon}{\sqrt{\epsilon}}\right)$ to obtain a nearly optimal low-rank approximation.

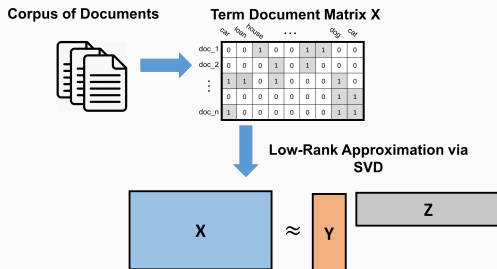
What do you think a stochastic version of Krylov subspace method would look like?

$$\mathcal{K}_q = \left[\mathbf{g}, (\mathbf{X}^T \mathbf{X}) \cdot \mathbf{g}, (\mathbf{X}^T \mathbf{X})^2 \cdot \mathbf{g}, \dots, (\mathbf{X}^T \mathbf{X})^q \cdot \mathbf{g} \right]$$

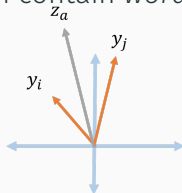
Applications of (partial) singular value decomposition:

- Low-rank approximation (data compression)
- Denoising, in-painting, matrix completion
- Semantic embeddings

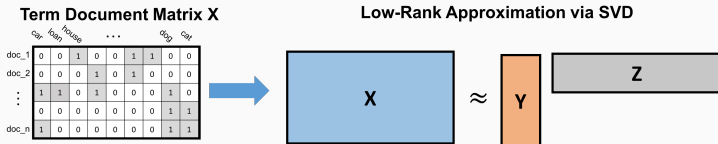
EXAMPLE: LATENT SEMANTIC ANALYSIS



- $\langle \vec{y}_i, \vec{z}_a \rangle \approx 1$ when doc_i contains $word_a$.
- If doc_i and doc_j both contain $word_a$, $\langle \vec{y}_i, \vec{z}_a \rangle \approx \langle \vec{y}_j, \vec{z}_a \rangle = 1$.



EXAMPLE: LATENT SEMANTIC ANALYSIS



- The columns $\vec{z}_1, \vec{z}_2, \dots$ give representations of words, with $\vec{z}_i$ and $\vec{z}_j$ tending to have high dot product if $word_i$ and $word_j$ appear in many of the same documents.
- Z corresponds to the top k right singular vectors: the eigenvectors of XX^T . Intuitively, what is XX^T ?
- $(XX^T)_{i,j} =$

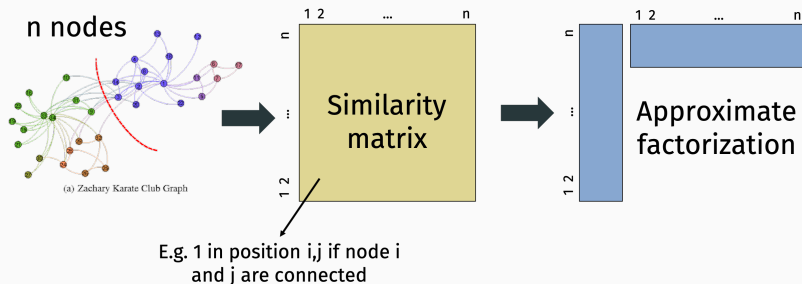
EXAMPLE: WORD EMBEDDING

Not obvious how to convert a word into a feature vector that captures the meaning of that word. Approach suggested by LSA: build a $d \times d$ symmetric “similarity matrix” $\mathbf{M}$ between words, and factorize: $\mathbf{M} \approx \mathbf{F}\mathbf{F}^T$ for rank k $\mathbf{F}$.

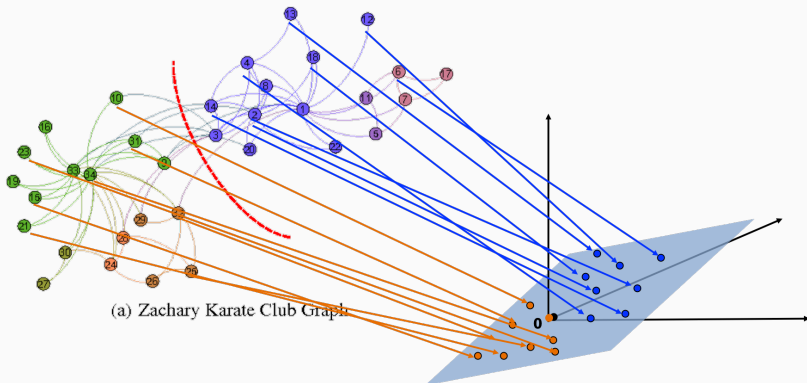
- **Similarity measures:** How often do $word_i, word_j$ appear in the same sentence, in the same window of w words, in similar positions of documents in different languages?
- Replacing $\mathbf{X}\mathbf{X}^T$ with these different metrics (sometimes appropriately transformed) leads to popular word embedding algorithms: **word2vec**, **GloVe**, etc.

ENCODING GRAPH SIMILARITY

Often data is represented as a graph and similarities can be obtained from that graph:



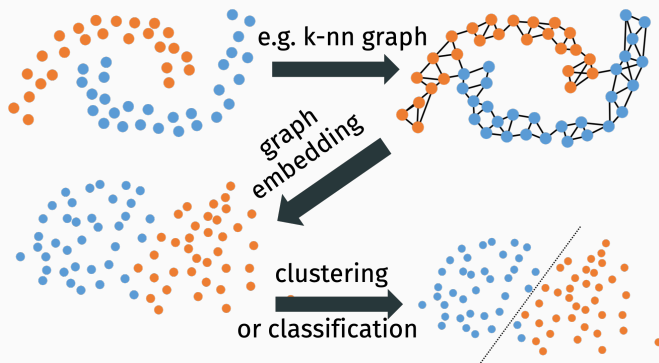
ENCODING GRAPH SIMILARITY



Social networks in 1970: “The network captures 34 members of a karate club, documenting links between pairs of members who interacted outside the club. During the study a conflict arose between the administrator “John A” and instructor “Mr. Hi” (pseudonyms), which led to the split of the club into two. Half of the members formed a new club around Mr. Hi; members from the other part found a new instructor or gave up karate. Based on collected data Zachary correctly assigned all but one member of the club to the groups they actually joined after the split.” – Wikipedia

SPECTRAL CLUSTERING

Idea: Construct synthetic graph for data that is hard to cluster.

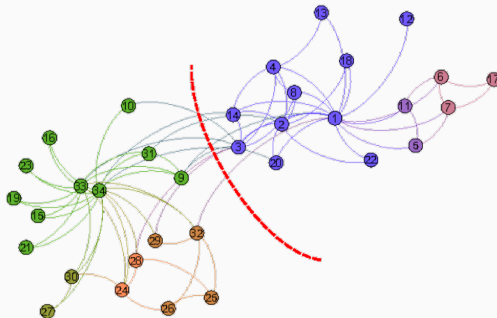


Spectral Clustering, Laplacian Eigenmaps, Locally linear embedding, Isomap, etc.

Spectral graph theory lets us formalize these heuristic ideas.

Goal: Partition nodes along a cut that:

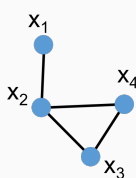
- Has few crossing edges: $|\{(u, v) \in E : u \in S, v \in T\}|$ is small.
- Separates large partitions: $|S|, |T|$ are not too small.



(a) Zachary Karate Club Graph

THE LAPLACIAN VIEW

For a graph with adjacency matrix A and degree matrix D ,
 $L = D - A$ is the **graph Laplacian**.


$$\overset{D}{\begin{bmatrix} 1 & 0 & 0 & 0 \\ 0 & 3 & 0 & 0 \\ 0 & 0 & 2 & 0 \\ 0 & 0 & 0 & 2 \end{bmatrix}} - \overset{A}{\begin{bmatrix} 0 & 1 & 0 & 0 \\ 1 & 0 & 1 & 1 \\ 0 & 1 & 0 & 1 \\ 0 & 1 & 1 & 0 \end{bmatrix}} = \overset{L}{\begin{bmatrix} 1 & -1 & 0 & 0 \\ -1 & 3 & -1 & -1 \\ 0 & -1 & 2 & -1 \\ 0 & -1 & -1 & 2 \end{bmatrix}}$$

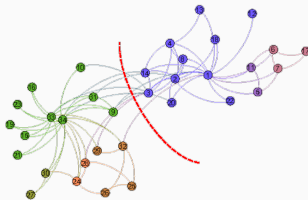
$L = B^T B$ where B is the “edge-vertex incidence” matrix.

$B =$

Conclusions from $L = B^T B$

- L is positive semidefinite: $\mathbf{x}^T L \mathbf{x} \geq 0$ for all $\mathbf{x}$.
- $L = V \Sigma^2 V^T$ where $U \Sigma^2 V^T$ is B 's SVD. Columns of V are eigenvectors of L .
- For a cut indicator vector $\mathbf{c} \in \{-1, 1\}^n$ with $\mathbf{c}(i) = -1$ for $i \in S$ and $\mathbf{c}(i) = 1$ for $i \in T$:
 - $\mathbf{c}^T L \mathbf{c} = \sum_{(i,j) \in E} (\mathbf{c}(i) - \mathbf{c}(j))^2 = 4 \cdot \text{cut}(S, T).$

THE LAPLACIAN VIEW



(a) Zachary Karate Club Graph

For a cut indicator vector $\mathbf{c} \in \{-1, 1\}^n$ with $\mathbf{c}(i) = -1$ for $i \in S$ and $\mathbf{c}(i) = 1$ for $i \in T$:

- $\mathbf{c}^T \mathbf{L} \mathbf{c} = 4 \cdot \text{cut}(S, T)$.
- $\mathbf{c}^T \mathbf{1} = |T| - |S|$.

Want to minimize both $\mathbf{c}^T \mathbf{L} \mathbf{c}$ (cut size) and $\mathbf{c}^T \mathbf{1}$ (imbalance).

Courant–Fischer min-max principle

Let $V = [\mathbf{v}_1, \dots, \mathbf{v}_n]$ be the eigenvectors of L .

$$\mathbf{v}_1 = \arg \max_{\|\mathbf{v}\|=1} \mathbf{v}^T L \mathbf{v}$$

$$\mathbf{v}_2 = \arg \max_{\|\mathbf{v}\|=1, \mathbf{v} \perp \mathbf{v}_1} \mathbf{v}^T L \mathbf{v}$$

$$\mathbf{v}_3 = \arg \max_{\|\mathbf{v}\|=1, \mathbf{v} \perp \mathbf{v}_1, \mathbf{v}_2} \mathbf{v}^T L \mathbf{v}$$

$$\vdots$$

$$\mathbf{v}_n = \arg \max_{\|\mathbf{v}\|=1, \mathbf{v} \perp \mathbf{v}_1, \dots, \mathbf{v}_{n-1}} \mathbf{v}^T L \mathbf{v}$$

Courant–Fischer min-max principle

Let $V = [\mathbf{v}_1, \dots, \mathbf{v}_n]$ be the eigenvectors of L .

$$\mathbf{v}_n = \arg \min_{\|\mathbf{v}\|=1, \mathbf{v}} \mathbf{v}^T L \mathbf{v}$$

$$\mathbf{v}_{n-1} = \arg \min_{\|\mathbf{v}\|=1, \mathbf{v} \perp \mathbf{v}_n} \mathbf{v}^T L \mathbf{v}$$

$$\mathbf{v}_{n-2} = \arg \min_{\|\mathbf{v}\|=1, \mathbf{v} \perp \mathbf{v}_n, \mathbf{v}_{n-1}} \mathbf{v}^T L \mathbf{v}$$

$$\vdots$$

$$\mathbf{v}_1 = \arg \min_{\|\mathbf{v}\|=1, \mathbf{v} \perp \mathbf{v}_n, \dots, \mathbf{v}_2} \mathbf{v}^T L \mathbf{v}$$

SMALLEST LAPLACIAN EIGENVECTOR

The smallest eigenvector/singular vector $\mathbf{v}_n$ satisfies:

$$\mathbf{v}_n = \frac{1}{\sqrt{n}} \cdot \mathbf{1} = \arg \min_{\mathbf{v} \in \mathbb{R}^n \text{ with } \|\mathbf{v}\|=1} \mathbf{v}^T L \mathbf{v}$$

with $\mathbf{v}_n^T L \mathbf{v}_n = 0$.

SECOND SMALLEST LAPLACIAN EIGENVECTOR

By Courant-Fischer, $\mathbf{v}_{n-1}$ is given by:

$$\mathbf{v}_{n-1} = \underset{\|\mathbf{v}\|=1, \mathbf{v}_n^T \mathbf{v}=0}{\operatorname{arg\,min}} \mathbf{v}^T L \mathbf{v}$$

If $\mathbf{v}_{n-1}$ were binary $\{-1, 1\}^n$ it would have:

- $\mathbf{v}_{n-1}^T L \mathbf{v}_{n-1} = \operatorname{cut}(S, T)$ as small as possible **given that**
 $\mathbf{v}_{n-1}^T \mathbf{1} = |T| - |S| = 0$.
- $\mathbf{v}_{n-1}$ would indicate the smallest perfectly balanced cut.

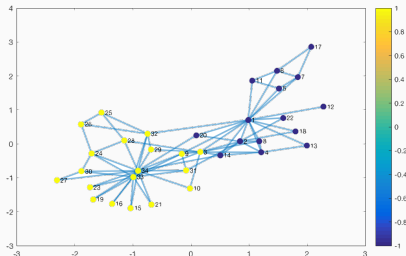
$\mathbf{v}_{n-1} \in \mathbb{R}^n$ is not generally binary, but still satisfies a ‘relaxed’ version of this property.

CUTTING WITH THE SECOND LAPLACIAN EIGENVECTOR

Find a good partition of the graph by computing

$$\mathbf{v}_{n-1} = \underset{\mathbf{v} \in \mathbb{R}^n \text{ with } \|\mathbf{v}\|=1, \mathbf{v}^T \mathbf{1}=0}{\operatorname{argmin}} \quad \mathbf{v}^T \mathbf{L} \mathbf{v}$$

Set S to be all nodes with $\mathbf{v}_{n-1}(i) < 0$, and T to be all with $\mathbf{v}_{n-1}(i) \geq 0$.

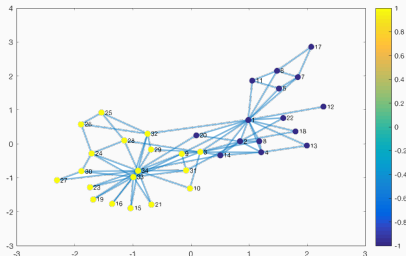


CUTTING WITH THE SECOND LAPLACIAN EIGENVECTOR

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$$\mathbf{v}_{n-1} = \underset{\mathbf{v} \in \mathbb{R}^n \text{ with } \|\mathbf{v}\|=1, \mathbf{v}^T \mathbf{1}=0}{\operatorname{arg\,min}} \quad \mathbf{v}^T \mathbf{L} \mathbf{v}$$

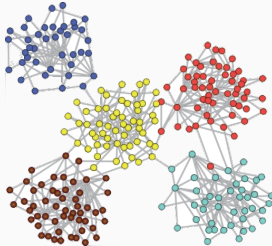
Set S to be all nodes with $\mathbf{v}_{n-1}(i) < 0$, and T to be all with $\mathbf{v}_{n-1}(i) \geq 0$.



SPECTRAL PARTITIONING IN PRACTICE

The Shi-Malik normalized cuts algorithm is one of the most commonly used variants of this approach, using the normalized Laplacian $\bar{\mathbf{L}} = \mathbf{D}^{-1/2} \mathbf{L} \mathbf{D}^{-1/2}$.

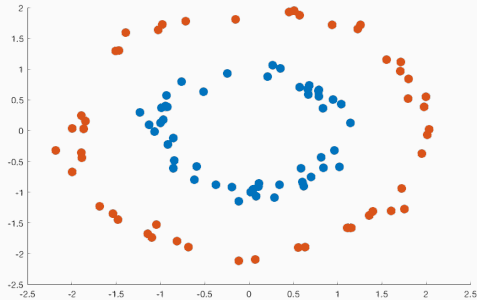
Important consideration: What to do when we want to split the graph into more than two parts?



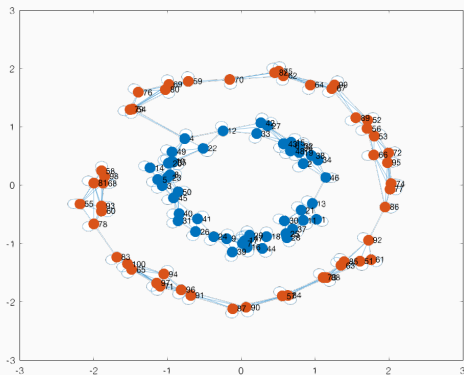
Spectral Clustering:

- Compute smallest k eigenvectors $\mathbf{v}_{n-1}, \dots, \mathbf{v}_{n-k}$ of $\mathbf{L}$.
- Represent each node by its corresponding row in $\mathbf{V} \in \mathbb{R}^{n \times k}$ whose rows are $\mathbf{v}_{n-1}, \dots, \mathbf{v}_{n-k}$.
- Cluster these rows using k -means clustering (or really any clustering method).

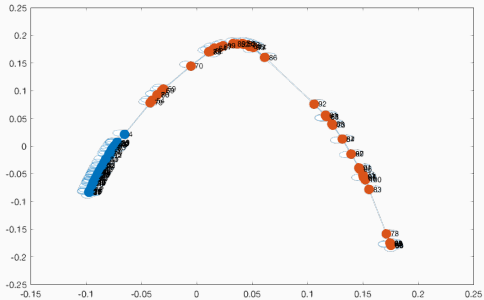
Original Data: (not linearly separable)



k -Nearest Neighbors Graph:



Embedding with eigenvectors $\mathbf{v}_{n-1}, \mathbf{v}_{n-2}$: (linearly separable)



So far: Showed that spectral clustering partitions a graph along a small cut between large pieces.

- No formal guarantee on the ‘quality’ of the partitioning.
- Would be difficult to analyze for general input graphs.

Common approach: Give a natural **generative model** for which produces random but realistic inputs and analyze how the algorithm performs on inputs drawn from this model.

- Very common in algorithm design for data analysis/machine learning (can be used to justify ℓ_2 linear regression, k -means clustering, PCA, etc.)

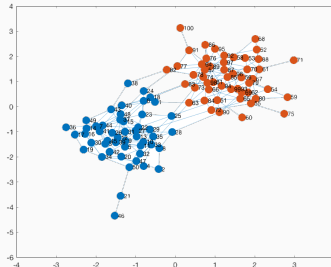
Ideas for a generative model for graphs that would allow us to understand partitioning?

STOCHASTIC BLOCK MODEL

Stochastic Block Model (Planted Partition Model):

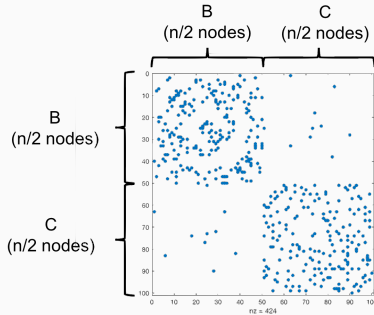
Let $G_n(p, q)$ be a distribution over graphs on n nodes, split equally into two groups B and C , each with $n/2$ nodes.

- Any two nodes in the **same group** are connected with probability p (including self-loops).
- Any two nodes in **different groups** are connected with prob. $q < p$.



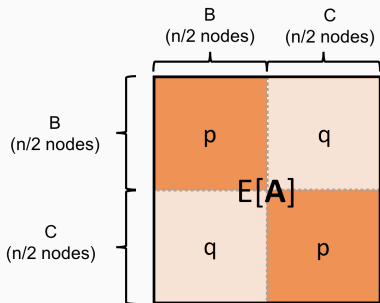
Let G be a stochastic block model graph drawn from $G_n(p, q)$.

- Let $\mathbf{A} \in \mathbb{R}^{n \times n}$ be the adjacency matrix of G . What is $\mathbb{E}[\mathbf{A}]$?



EXPECTED ADJACENCY SPECTRUM

Letting G be a stochastic block model graph drawn from $G_n(p, q)$ and $\mathbf{A} \in \mathbb{R}^{n \times n}$ be its adjacency matrix. $(\mathbb{E}[\mathbf{A}])_{i,j} = p$ for i, j in same group, $(\mathbb{E}[\mathbf{A}])_{i,j} = q$ otherwise.



What are the eigenvectors and eigenvalues of $\mathbb{E}[\mathbf{A}]$?

Letting G be a stochastic block model graph drawn from $G_n(p, q)$ and $\mathbf{A} \in \mathbb{R}^{n \times n}$ be its adjacency matrix, what are the eigenvectors and eigenvalues of $\mathbb{E}[\mathbf{A}]$?

EXPECTED ADJACENCY SPECTRUM

$$\begin{array}{c}
 \begin{array}{cc}
 \text{B} & \text{C} \\
 (n/2 \text{ nodes}) & (n/2 \text{ nodes})
 \end{array} \\
 \begin{array}{|c|c|}
 \hline
 \begin{array}{c} p \\ q \end{array} & \begin{array}{c} q \\ p \end{array} \\
 \hline
 \end{array}
 \end{array}
 \begin{array}{c}
 \text{E}[\mathbf{A}] \\
 \hline
 \end{array}
 =
 \begin{array}{c}
 \mathbf{V} \\
 \hline
 \begin{array}{|c|}
 \hline
 \begin{array}{c} 1 \ 1 \\ 1 \ 1 \\ 1 \ 1 \\ 1 \ 1 \\ 1 \ -1 \\ 1 \ -1 \\ 1 \ -1 \\ 1 \ -1 \end{array} \\
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 \begin{array}{c} \frac{n(p+q)}{2} \\ \frac{n(p-q)}{2} \end{array} \\
 \hline
 \end{array}
 \end{array}
 \begin{array}{c}
 \mathbf{V}^T \\
 \hline
 \begin{array}{|c|}
 \hline
 \begin{array}{cccccccc} 1 & 1 & 1 & 1 & 1 & 1 & 1 & 1 \\ 1 & 1 & 1 & 1 & -1 & -1 & -1 & -1 \end{array} \\
 \hline
 \end{array}
 \end{array}$$

- $\vec{v}_1 = \vec{1}$ with eigenvalue $\lambda_1 = \frac{(p+q)n}{2}$.
- $\vec{v}_2 = \chi_{B,C}$ with eigenvalue $\lambda_2 = \frac{(p-q)n}{2}$.
- $\chi_{B,C}(i) = 1$ if $i \in B$ and $\chi_{B,C}(i) = -1$ for $i \in C$.

If we compute $\vec{v}_2$ then we recover the communities B and C !

Letting G be a stochastic block model graph drawn from $G_n(p, q)$, $\mathbf{A} \in \mathbb{R}^{n \times n}$ be its adjacency matrix and $\mathbf{L}$ be its Laplacian, what are the eigenvectors and eigenvalues of $\mathbb{E}[\mathbf{L}]$?

Upshot: The second small eigenvector of $\mathbb{E}[\mathbf{L}]$ is $\chi_{B,C}$ – the indicator vector for the cut between the communities.

- If the random graph G (equivalently $\mathbf{A}$ and $\mathbf{L}$) were exactly equal to its expectation, partitioning using this eigenvector would exactly recover communities B and C .

How do we show that a matrix (e.g., $\mathbf{A}$) is close to its expectation? **Matrix concentration inequalities.**

- Analogous to scalar concentration inequalities like Markovs, Chebyshevs, Bernsteins.
- Random matrix theory is a very recent and cutting edge subfield of mathematics that is being actively applied in computer science, statistics, and machine learning.

Matrix Concentration Inequality: If $p \geq O\left(\frac{\log^4 n}{n}\right)$, then with high probability

$$\|\mathbf{A} - \mathbb{E}[\mathbf{A}]\|_2 \leq O(\sqrt{pn}).$$

where $\|\cdot\|_2$ is the matrix **spectral** norm (operator norm).

For $\mathbf{X} \in \mathbb{R}^{n \times d}$, $\|\mathbf{X}\|_2 = \max_{\mathbf{z} \in \mathbb{R}^d: \|\mathbf{z}\|_2=1} \|\mathbf{X}\mathbf{z}\|_2$.

Exercise: Show that $\|\mathbf{X}\|_2$ is equal to the largest singular value of $\mathbf{X}$. For symmetric $\mathbf{X}$ (like $\mathbf{A} - \mathbb{E}[\mathbf{A}]$) show that it is equal to the magnitude of the largest magnitude eigenvalue.

For the stochastic block model application, we want to show that the second eigenvectors of $\mathbf{A}$ and $\mathbb{E}[\mathbf{A}]$ are close. How does this relate to their difference in spectral norm?

Davis-Kahan Eigenvector Perturbation Theorem: Suppose $\mathbf{A}, \bar{\mathbf{A}} \in \mathbb{R}^{d \times d}$ are symmetric with $\|\mathbf{A} - \bar{\mathbf{A}}\|_2 \leq \epsilon$ and eigenvectors $v_1, v_2, \dots, v_d$ and $\bar{v}_1, \bar{v}_2, \dots, \bar{v}_d$. Letting $\theta(v_i, \bar{v}_i)$ denote the angle between v_i and $\bar{v}_i$, for all i :

$$\sin[\theta(v_i, \bar{v}_i)] \leq \frac{\epsilon}{\min_{j \neq i} |\lambda_i - \lambda_j|}$$

where $\lambda_1, \dots, \lambda_d$ are the eigenvalues of $\bar{\mathbf{A}}$.

The error gets larger if there are eigenvalues with similar magnitudes.

EIGENVECTOR PERTURBATION

$$\begin{array}{c} \mathbf{A} \\ \boxed{\begin{array}{cc} 1+\varepsilon & 0 \\ 0 & 1 \end{array}} \end{array} - \begin{array}{c} \bar{\mathbf{A}} \\ \boxed{\begin{array}{cc} 1 & 0 \\ 0 & 1+\varepsilon \end{array}} \end{array} = \begin{array}{c} \mathbf{A}-\bar{\mathbf{A}} \\ \boxed{\begin{array}{cc} \varepsilon & 0 \\ 0 & \varepsilon \end{array}} \end{array}$$

APPLICATION TO STOCHASTIC BLOCK MODEL

Claim 1 (Matrix Concentration): For $p \geq O\left(\frac{\log^4 n}{n}\right)$,

$$\|\mathbf{A} - \mathbb{E}[\mathbf{A}]\|_2 \leq O(\sqrt{pn}).$$

Claim 2 (Davis-Kahan): For $p \geq O\left(\frac{\log^4 n}{n}\right)$,

$$\sin \theta(v_2, \bar{v}_2) \leq \frac{O(\sqrt{pn})}{\min_{j \neq i} |\lambda_i - \lambda_j|} \leq \frac{O(\sqrt{pn})}{(p-q)n/2} = o\left(\frac{\sqrt{p}}{(p-q)\sqrt{n}}\right)$$

Recall: $\mathbb{E}[\mathbf{A}]$, has eigenvalues $\lambda_1 = \frac{(p+q)n}{2}$, $\lambda_2 = \frac{(p-q)n}{2}$, $\lambda_i = 0$ for $i \geq 3$.

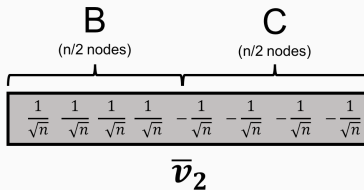
$$\min_{j \neq i} |\lambda_i - \lambda_j| = \min\left(qn, \frac{(p-q)n}{2}\right).$$

Typically, $\frac{(p-q)n}{2}$ will be the minimum of these two gaps.

APPLICATION TO STOCHASTIC BLOCK MODEL

So Far: $\sin \theta(v_2, \bar{v}_2) \leq O\left(\frac{\sqrt{p}}{(p-q)\sqrt{n}}\right)$. What does this give us?

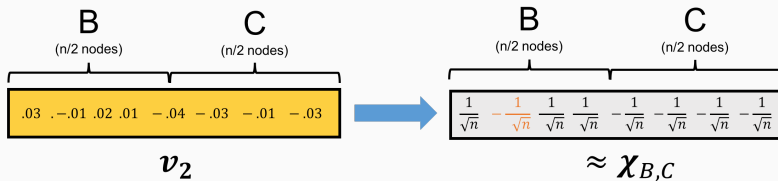
- Can show that this implies $\|v_2 - \bar{v}_2\|_2^2 \leq O\left(\frac{p}{(p-q)^2 n}\right)$ (exercise).
- $\bar{v}_2$ is $\frac{1}{\sqrt{n}}\chi_{B,C}$: the community indicator vector.



- Every i where $v_2(i)$, $\bar{v}_2(i)$ differ in sign contributes $\geq \frac{1}{n}$ to $\|v_2 - \bar{v}_2\|_2^2$.
- So they differ in sign in at most $O\left(\frac{p}{(p-q)^2}\right)$ positions.

APPLICATION TO STOCHASTIC BLOCK MODEL

Upshot: If G is a stochastic block model graph with adjacency matrix A , if we compute its second large eigenvector v_2 and assign nodes to communities according to the sign pattern of this vector, we will correctly assign all but $O\left(\frac{p}{(p-q)^2}\right)$ nodes.



- Why does the error increase as q gets close to p ?
- Even when $p - q = O(1/\sqrt{n})$, assign all but an $O(n)$ fraction of nodes correctly. E.g., assign 99% of nodes correctly.